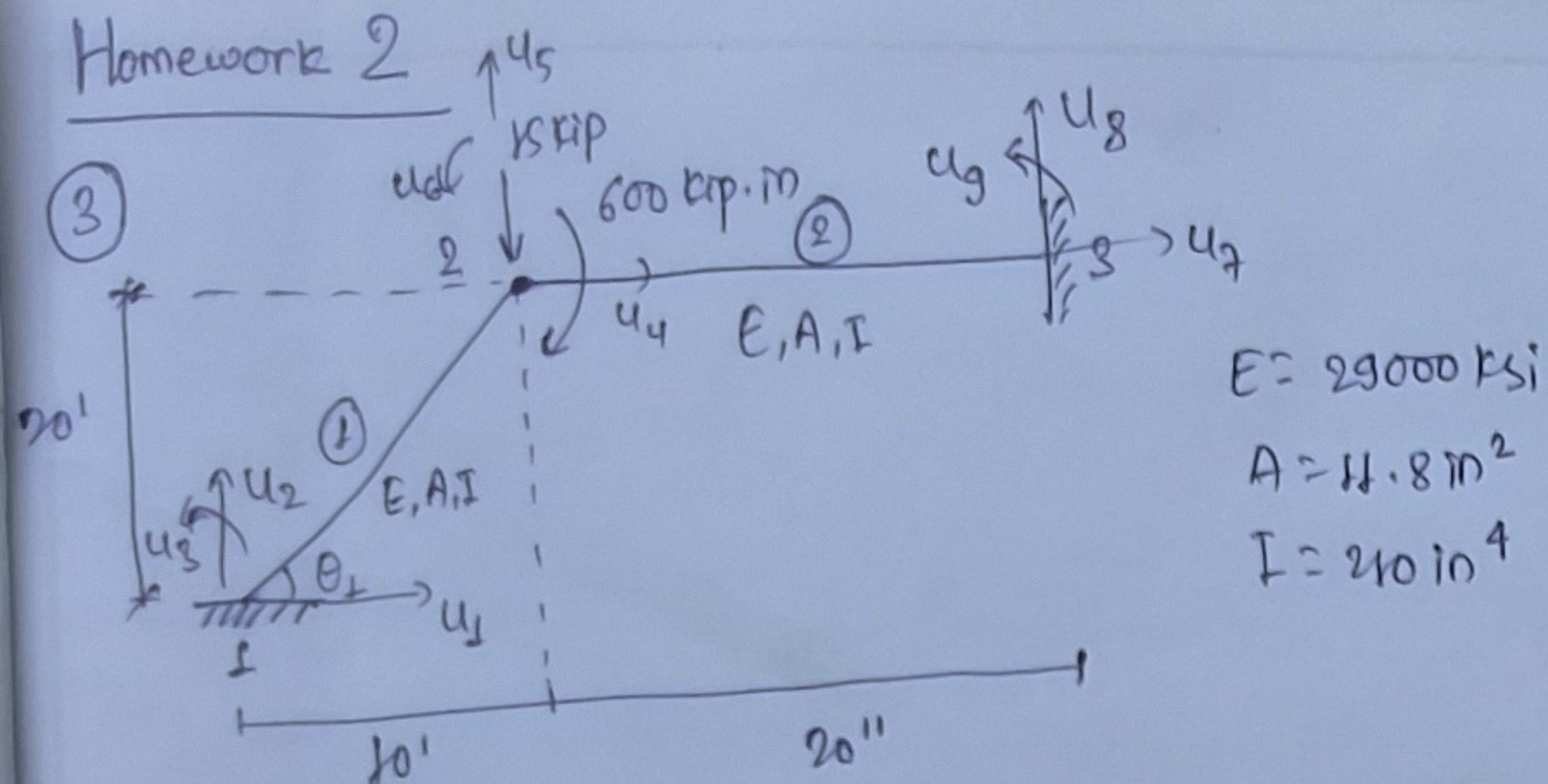


Homework 2



Solution.

for element ①, $\theta_1 = \tan^{-1}(20/10) = 63.43^\circ$
 $C = \cos \theta = 0.447, \sin \theta = 0.894 = S$

$$[T] = \begin{bmatrix} C & S & 0 & 0 & 0 & 0 \\ -S & C & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & C & S & 0 \\ 0 & 0 & 0 & -S & C & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

$$[T]_1 = \begin{bmatrix} 0.447 & 0.894 & 0 & 0 & 0 & 0 \\ -0.894 & 0.447 & 0 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0.447 & 0.894 & 0 \\ 0 & 0 & 0 & -0.894 & 0.447 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

$$K_{\text{local}} = \begin{bmatrix} AE/l & 0 & 0 & -AE/l & 0 & 0 \\ 0 & 12EI/l^3 & 6EI/l^2 & 0 & -12EI/l^3 & 6EI/l^2 \\ 0 & 6EI/l^2 & 4EI/l & 0 & -6EI/l^2 & 2EI/l \\ -AE/l & 0 & 0 & AE/l & 0 & 0 \\ 0 & -12EI/l^3 & -6EI/l^2 & 0 & 12EI/l^3 & -6EI/l^2 \\ 0 & 6EI/l^2 & 2EI/l & 0 & -6EI/l^2 & 4EI/l \end{bmatrix}$$

$$K_{\text{global}} = [T]^T K_{\text{local}} [T]$$

Calculated directly using python,

$$[K]_{\text{global}(1)} = 10^4 \begin{bmatrix} 0.829 & 0.350 & -6.536 & -0.829 & -0.350 & -6.536 \\ 0.350 & 1.355 & 3.268 & -0.350 & -1.355 & 3.268 \\ -6.536 & 3.268 & 108.9 & 6.536 & -3.268 & 54.471 \\ -0.829 & -0.350 & 6.536 & 0.829 & 0.350 & 6.536 \\ -0.350 & -1.355 & -3.268 & 0.350 & 1.355 & -3.268 \\ -6.536 & 3.268 & 54.471 & 6.536 & -3.268 & 108.9 \end{bmatrix}$$

Similarly for element ②, $\theta_2 = 0^\circ \Rightarrow \cos \theta = 1, \sin \theta = 0$,

$$[K]_{\text{global}(2)} = 10^4 \begin{bmatrix} 1.711 & 0 & 0 & -1.711 & 0 & 0 \\ 0 & 0.913 & 9.135 & 0 & -0.913 & 9.135 \\ 0 & 9.135 & 91.8 & 0 & -9.135 & 91.8 \\ 1.711 & 0 & 0 & 1.711 & 0 & 0 \\ 0 & -0.913 & -9.135 & 0 & 0.913 & -9.135 \\ 0 & 9.135 & 91.8 & 0 & -9.135 & 91.8 \end{bmatrix}$$

Symmetric

Superposition of the stiffness gives stiffness matrix for the structure,

$$[K] = \begin{bmatrix} 0.829 & 0.350 & -6.536 & -0.829 & -0.350 & -6.536 & 0 & 0 & 0 \\ 0.350 & 1.355 & 3.268 & -0.350 & -1.355 & 3.268 & 0 & 0 & 0 \\ -6.536 & 3.268 & 108.9 & 6.536 & -3.268 & 54.471 & 0 & 0 & 0 \\ -0.829 & -0.350 & 6.536 & 0.829 & 0.350 & 6.536 & 1.711 & 0 & 0 \\ -0.350 & -1.355 & -3.268 & 0.350 & 1.355 & -3.268 & 0 & 0.913 & -9.135 \\ -6.536 & 3.268 & 54.471 & 6.536 & -3.268 & 108.9 & 0 & -9.135 & 91.8 \\ 0 & 0 & 0 & 1.711 & 0 & 0 & 1.711 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0.913 & 9.135 & 0 & 0.913 & -9.135 \\ 0 & 0 & 0 & 0 & -9.135 & 91.8 & 0 & -9.135 & 91.8 \end{bmatrix}$$

Symmetric

$$\{F\} = [K] \{u\}$$

$$\{F\} = \begin{Bmatrix} f_1 \\ f_2 \\ m_3 \\ 0 \\ -15 \\ -600 \\ f_7 \\ f_8 \\ m_9 \end{Bmatrix}$$

$$\{u\} = \begin{Bmatrix} 0 \\ 0 \\ 0 \\ u_4 \\ u_5 \\ u_6 \\ 0 \\ 0 \\ 0 \end{Bmatrix}$$

To solve for displacements,

$$\begin{Bmatrix} f_4 \\ f_5 \\ m_6 \end{Bmatrix} = \begin{Bmatrix} 0 \\ -15 \\ -600 \end{Bmatrix} = \begin{bmatrix} 2.54 & 0.350 & 6.536 \\ 0.350 & 2.268 & 5.867 \\ 6.536 & 5.867 & 230.7 \end{bmatrix} \begin{Bmatrix} u_4 \\ u_5 \\ u_6 \end{Bmatrix}$$

$$\Rightarrow \begin{Bmatrix} u_4 \\ u_5 \\ u_6 \end{Bmatrix} = \begin{Bmatrix} 7.258 \\ -0.507 \\ -2.793 \end{Bmatrix} = \{d\} \text{ say}$$

Now,

$$f_1 = \begin{bmatrix} -0.629 & -0.350 & -6.536 \end{bmatrix} \begin{Bmatrix} 7.258 \\ -0.507 \\ -2.793 \end{Bmatrix} = 12.415 \text{ kip}$$

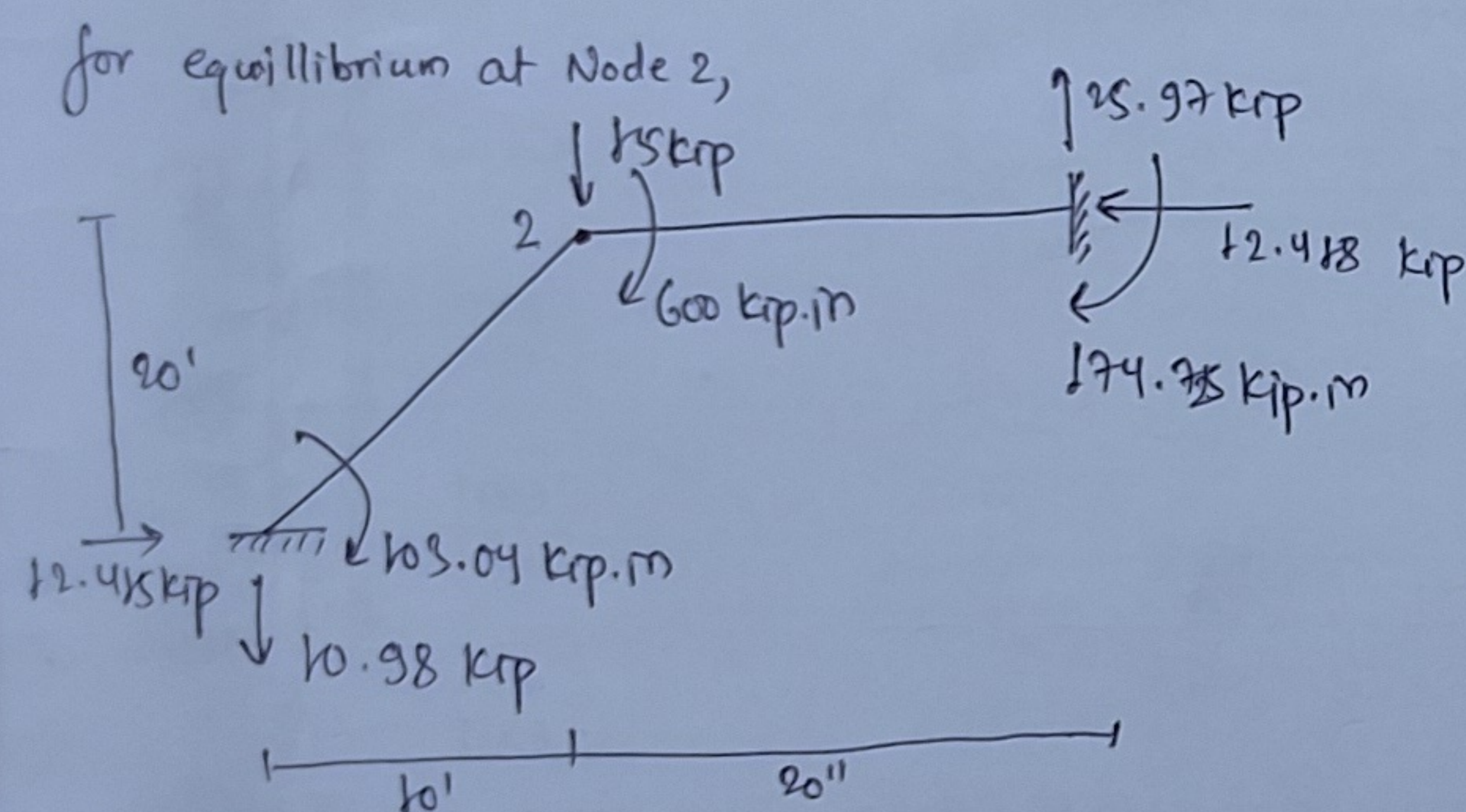
$$f_2 = \begin{bmatrix} -0.350 & -1.355 & 3.268 \end{bmatrix} \begin{Bmatrix} d \end{Bmatrix} = -10.98 \text{ kip}$$

$$m_3 = \begin{bmatrix} 6.536 & -3.268 & 54.471 \end{bmatrix} \begin{Bmatrix} d \end{Bmatrix} = -103.04 \text{ kip}\cdot\text{m}$$

$$f_7 = \begin{bmatrix} -1.711 & 0 & 0 \end{bmatrix} \begin{Bmatrix} d \end{Bmatrix} = -12.418 \text{ kip}$$

$$f_8 = \begin{bmatrix} 0 & -0.913 & -9.135 \end{bmatrix} \begin{Bmatrix} d \end{Bmatrix} = 25.97 \text{ kip}$$

$$m_9 = \begin{bmatrix} 0 & 9.135 & 60.9 \end{bmatrix} \begin{Bmatrix} d \end{Bmatrix} = -174.725 \text{ kip}\cdot\text{m}$$



$$\sum M_2 = 0 = 600 + 174.725 + 103.04$$

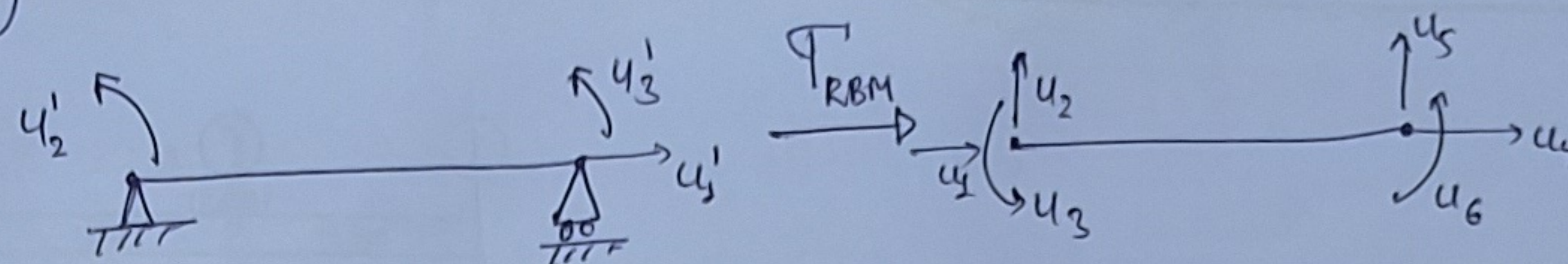
$$- 12.415 \times 20$$

$$- 10.98 \times 10$$

$$+ 25.97 \times 20$$

$$= 0.265 \approx 0$$

①



Solⁿ:

Axial Behaviour (axial elongation u_1')

governing eqⁿ, $\frac{d^2 u(x)}{dx^2} = 0$ (for constant EA)

Integrating twice, $u(x) = C_1 x + C_2$

Boundary Conditions,

At $x=0$, $u=0 \Rightarrow C_2=0$

At $x=L$, $u=u_3' \Rightarrow u_3' = C_1 L \Rightarrow C_1 = u_3'/L$

$$\Rightarrow u(x) = \frac{u_3'}{L} x$$

$$E = \frac{\text{Stress}}{\text{Strain}} \Rightarrow E = \frac{f_{ax}/A}{\frac{du(x)}{dx}}$$

$$\Rightarrow f_{ax}' = AE \frac{du(x)}{dx} = \left(\frac{AE}{L}\right) u_3'$$

Bending Behaviour (Rotations u_2' and u_4')

governing eqⁿ for constant EI and nodal loads only,

$$\frac{d^4 y(x)}{dx^4} = 0$$

$$\text{Integrating four times, } y'(x) = C_3 x^3 + C_4 x^2 + C_5 x + C_6 \quad \text{--- (1)}$$

$$y(x) = \frac{dy(x)}{dx} = 3C_3 x^2 + 2C_4 x + C_5$$

Boundary Conditions,

At $x=0$,

$$y'=0 \Rightarrow C_6=0 \quad \text{--- (a)}$$

$$y=u_2' \Rightarrow C_5=u_2' \quad \text{--- (b)}$$

At $x=L$,

$$y'=0 \Rightarrow 0 = C_3 L^2 + C_4 L + C_5 \Rightarrow C_3 L^2 + C_4 L + u_2' = 0$$

$$y=u_4' \Rightarrow u_4' = 3C_3 L^2 + 2C_4 L + u_2'$$

$$\Rightarrow u_3' - u_2' = 3(-u_2' - C_4 L) + 2C_4 L$$

$$\Rightarrow u_3' - u_2' = -3u_2' - 3C_4 L + 2C_4 L$$

$$\Rightarrow u_3' + 2u_2' = -C_4 L$$

$$\Rightarrow C_4 = -\frac{(2u_2' + u_3')}{L} \quad \text{--- (c)}$$

$$\Rightarrow C_3 = \frac{1}{L^2} (-u_2' + 2u_2' + u_3')$$

$$C_3 = \frac{u_2' + u_3'}{L^2} \quad \text{--- (d)}$$

$$\Rightarrow y'(x) = \left(\frac{u_2' + u_3'}{L^2}\right) x^3 - \frac{(2u_2' + u_3')}{L} x^2 + u_2' x \quad \text{--- (2)}$$

$$M(x) = EI \frac{d^2 y}{dx^2}$$

$$M'(x) = \left(\frac{u_2' + u_3'}{L^2}\right) EI (6x) - \frac{(2u_2' + u_3')}{L} 2EI \quad \text{--- (3)}$$

At $x=0$,

$$m_2' = -M = \frac{EI}{L} (4u_2' + 2u_3')$$

At $x=L$,

$$m_3' = M = EI (6L) \frac{u_2' + u_3'}{L^2} - 2EI \left(\frac{2u_2' + u_3'}{L}\right)$$

$$= \frac{EI}{L} (6u_2' + 6u_3' - 4u_2' - 2u_3')$$

$$= \frac{EI}{L} (2u_2' + 4u_3')$$

Basic Stiffness Matrix,

$$[K]_{\text{base}} = \begin{bmatrix} \frac{AE}{L} & 0 & 0 \\ 0 & 4EI/L & 2EI/L \\ 0 & 2EI/L & 4EI/L \end{bmatrix}$$

$$[T]_{\text{RBM}} = \begin{bmatrix} -1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1/L & 1 & 0 & -1/L & 0 \\ 0 & 1/L & 0 & 0 & -1/L & 1 \end{bmatrix}$$

$$[K]_{\text{global}} = T_{\text{RBM}}^T K_{\text{basic}} T_{\text{RBM}}$$

$$K_{\text{basic}} T_{\text{RBM}} = \begin{bmatrix} AE/L & 0 & 0 \\ 0 & 4EI/L & 2EI/L \\ 0 & 2EI/L & 4EI/L \end{bmatrix} \begin{bmatrix} -1 & 0 & 0 & 1 & 0 & 0 \\ 0 & 1/L & 1 & 0 & -1/L & 0 \\ 0 & 1/L & 0 & 0 & -1/L & 1 \end{bmatrix}$$

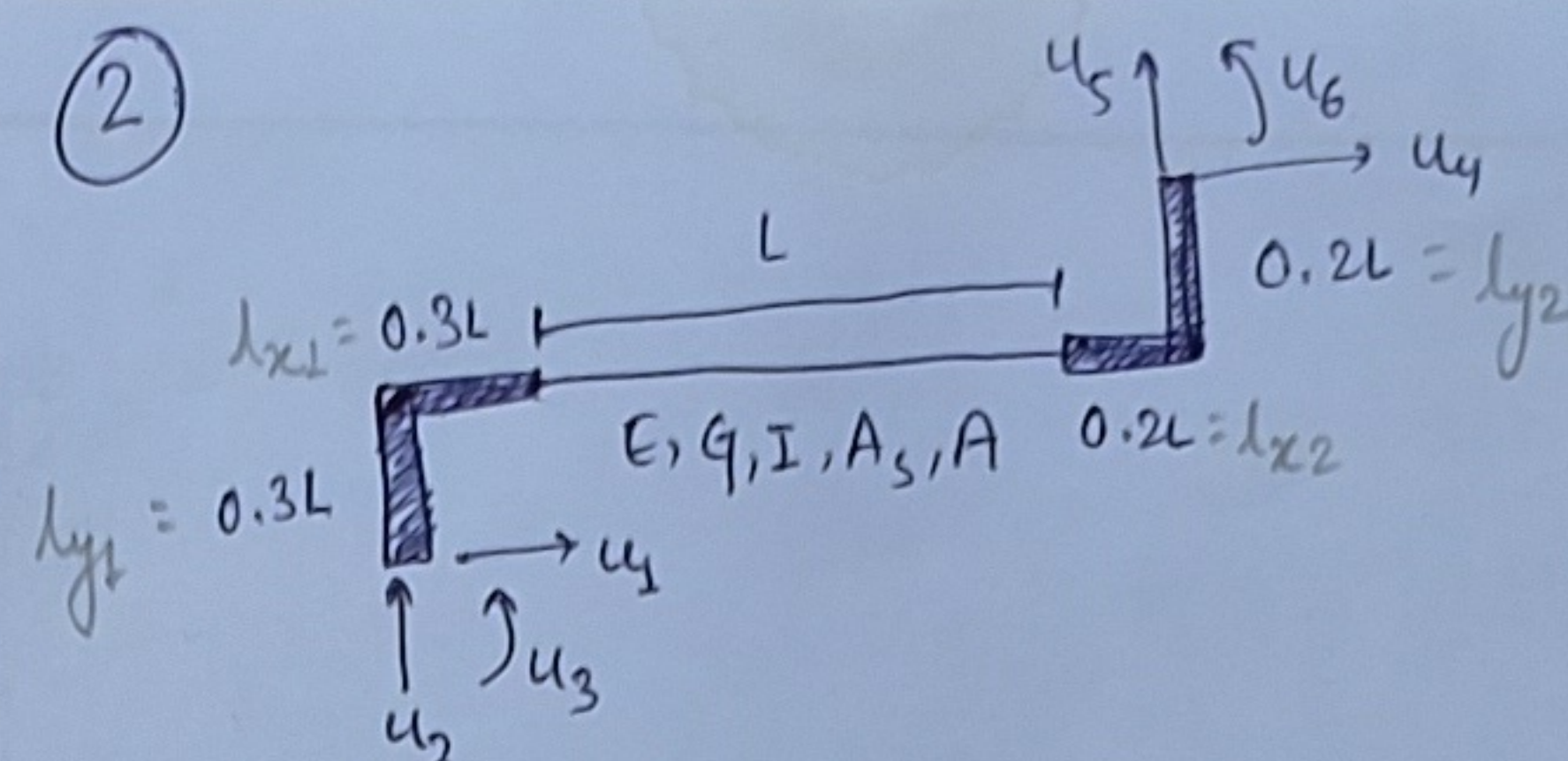
$$= \begin{bmatrix} -AE/L & 0 & 0 & AE/L & 0 & 0 \\ 0 & 6EI/L^2 & 4EI/L & 0 & -6EI/L^2 & 2EI/L \\ 0 & 6EI/L^2 & 4EI/L & 0 & -6EI/L^2 & 4EI/L \end{bmatrix}$$

$$T_{\text{RBM}}^T = \begin{bmatrix} -1 & 0 & 0 \\ 0 & 1/L & 1/L \\ 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & -1/L & -1/L \\ 0 & 0 & 1 \end{bmatrix}$$

$$[K]_{\text{global}} = \begin{bmatrix} AE/L & 0 & 0 & -AE/L & 0 & 0 \\ 0 & 12EI/L^3 & 6EI/L^2 & 0 & -12EI/L^3 & 6EI/L^2 \\ 0 & 6EI/L^2 & 4EI/L & 0 & -6EI/L^2 & 2EI/L \\ -AE/L & 0 & 0 & AE/L & 0 & 0 \\ 0 & -12EI/L^3 & -6EI/L^2 & 0 & 12EI/L^3 & -6EI/L^2 \\ 0 & 6EI/L^2 & 2EI/L & 0 & -6EI/L^2 & 4EI/L \end{bmatrix}$$

This is the same stiffness matrix derived in HW1, Question 4.

(2)



→ Linear-elastic Timoshenko beam theory
→ To derive 6x6 element stiffness matrix

Solⁿ:

from class,

$$[K]_{\text{local}} = \frac{EI}{L(1+\phi)} \begin{bmatrix} 12/L^2 & 6/L & -12/L^2 & 6/L \\ 6/L & 4+\phi & -6/L & 2-\phi \\ -12/L^2 & -6/L & 12/L^2 & -6/L \\ 6/L & 2-\phi & -6/L & 4+\phi \end{bmatrix}$$

$$[K]_{\text{global}} = T_{\text{REZ}}^T [K]_{\text{local}} T_{\text{REZ}}$$

$$T_{\text{REZ}} = \begin{bmatrix} 1 & 0 & -ly_1 & 0 & 0 & 0 \\ 0 & 1 & lx_1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & -ly_2 \\ 0 & 0 & 0 & 0 & 1 & lx_2 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & -0.3L & 0 & 0 & 0 \\ 0 & 1 & 0.3L & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & -0.2L \\ 0 & 0 & 0 & 0 & 1 & 0.2L \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

$$[K]_{\text{local}} T_{\text{REZ}} = \frac{EI}{L(1+\phi)} \begin{bmatrix} A(1+\phi)/I & 0 & -0.3A(1+\phi)L/I & -A(1+\phi)/I & 0 & 0.2A(1+\phi)L/I \\ 0 & 12/L^2 & 9.6/L & 0 & -12/L^2 & 3.6/L \\ 0 & 6/L & 5.8+\phi & 0 & -6/L & 0.8-\phi \\ -A(1+\phi)/I & 0 & 0.3A(1+\phi)L/I & A(1+\phi)/I & 0 & -0.2A(1+\phi)L/I \\ 0 & -12/L^2 & -9.6/L & 0 & 12/L^2 & -3.6/L \\ 0 & 6/L & 3.8-\phi & 0 & -6/L & 2.8+\phi \end{bmatrix}$$

$$T_{\text{REZ}}^T [K]_{\text{local}} T_{\text{REZ}} = \frac{EI}{L(1+\phi)} \begin{bmatrix} A(1+\phi)/I & 0 & -0.3AL(1+\phi)/I & -A(1+\phi)/I & 0 & 0.2AL(1+\phi)/I \\ 0 & 12/L^2 & 9.6/L & 0 & -12/L^2 & 3.6/L \\ -0.3AL(1+\phi)/I & 9.6/L & \frac{0.09A^2L^2(1+\phi)}{I} + 8.68 & \frac{0.3AL(1+\phi)}{I} & -9.6/L & -\frac{0.06AL^2(1+\phi)}{I} + 1.88-\phi \\ -A(1+\phi)/I & 0 & \frac{0.3AL(1+\phi)}{I} & \frac{A(1+\phi)}{I} & 0 & -\frac{0.2AL(1+\phi)}{I} \end{bmatrix}$$

